Topic 2 Review [100 marks]

1. [Maximum mark: 4] 25M.1.SL.TZ2.4

Construct a logic diagram for the following expression:

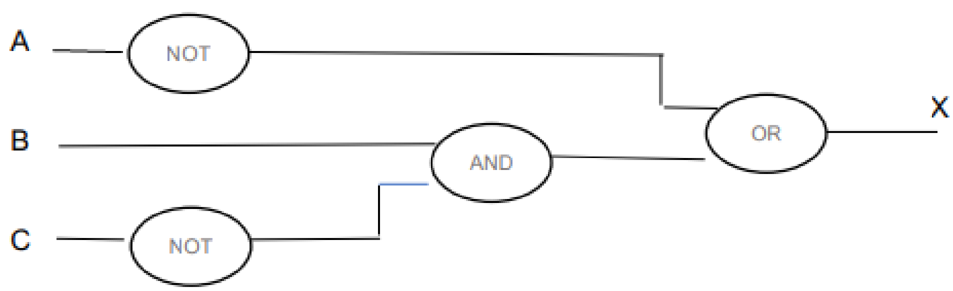
X = NOT A OR B AND NOT C [4]

Markscheme

*Award **max** [4].*

Award [1] for each of the correctly placed logic gates.

Note: *Each logic gate needs to have the correct input(s) and output.*



Examiners report

One of the most common mistakes in this question was that students created a truth table instead of constructing a logic diagram, resulting in the loss of all four marks. Those who drew the circuit diagram failed to consider the precedence of the logic operators, which led to them earning only two out of four marks.

2. [Maximum mark: 2] 25M.1.SL.TZ2.8

A binary number is held in the following 12-bit register:

1	1	1	1	0	0	1	1	1	0	1	0

(a) State the hexadecimal representation of this binary number. [1]

Markscheme

*Award **max** [1].*

F3A;

Examiners report

A vast majority of the students achieved a mark here.

(b) State how many different binary numbers can be represented in a 12-bit register. [1]

Markscheme

*Award **max** [1].*

2^{12}/4096;

Examiners report

Few students achieved a mark here.

3. [Maximum mark: 4] 25M.1.HL.TZ2.2

(a) Outline the purpose of secondary memory. [2]

Markscheme

*Award **max** [2].*

Award [1] for a function/purpose and [1] for a reasonable expansion.

to store data permanently (even if the power supply is off);
So, it can be accessed at a later time;

to provide large storage;
can store large files (like videos, images, audios, files, etc.) permanently;

to store/transfer data from one computer to another;
removable secondary storage devices can be easily disconnected and used on different systems;

to back up important data;
data backups used for disaster recovery due to system failure/accidental deletion;

to retain data;
beyond the capacity constraints of primary storage;

to enhance system performance/allow the system to handle more tasks simultaneously;
by swapping data between RAM/primary memory and secondary memory/Virtual memory;

to offer remote/online storage (cloud storage);
enabling users to access data/files from any device/collaborate easily across locations;

to store large databases (for businesses/science/research/education);
supporting data retrieval/management;

Examiners report

Many students described what secondary memory is, stated the characteristics of it and provided examples, rather than focusing on explaining its purpose in a computer.

(b) Describe a situation when secondary memory would be used as an extension of primary memory. [2]

Markscheme

*Award **max** [2]*

Primary memory required by all running processes exceeds the volume of RAM installed (shortage of physical memory);

A program currently used is too large to fit into the RAM (*Accept a description, such as attempting to play a video game which is much bigger in size than the RAM, when editing a large video file in specialised software*);

Part of the program (inactive pages) that is not currently being used can be stored in secondary memory temporarily/until it is required;

Note: *Terms like virtual memory, segmentation, paging may be used.*

Examiners report

Many students recognized that secondary memory would be used as an extension of primary memory when primary memory required by all running processes exceeds the volume of RAM installed. A program currently used is too large to fit into the RAM, so part of the program that is not currently being used can be stored in secondary memory temporarily. Some students in their descriptions correctly used terms like virtual memory, segmentation and paging.

4. [Maximum mark: 4] 25M.1.HL.TZ2.3

Outline **two** possible hardware upgrades to improve a desktop computer's performance. [4]

Markscheme

*Award **max** [4].*

*Award [1] for a hardware upgrade and [1] for an expansion/description of improvement, **x2**.*

Upgrade processor (resulting in faster processor or more cores);
To increase the speed of execution/efficiency of the computer/make the browser more responsive;

Add more RAM;
allows faster access to/use of memory-intensive applications (*Accept examples such as graphic software, etc.*);

Upgrade storage/increase the hard disk/SSD size;
get a faster drive with more storage space/more reliable storage devices;

Replace the graphic card with a new one;
So that colourful images on the monitor are of better quality;

Upgrade peripherals;
to have full support for the electronic devices/to increase efficiency of the computer;

Note:
Reward other reasonable answers such as upgrade CPU cooling system (which prevents overheating, allowing components to run at optimal speeds), upgrade expansion cards (Network adapter, Wireless network adapter and so on).

Accept examples, such as:

Upgrade I/O devices (keyboard/mice/scanner/display, etc.);
newer/faster/efficient I/O devices can impact the overall performance of the computer;

replace the old monitor with a new one;
to improve display quality (colour accuracy, etc.).

Examiners report

Many good and comprehensive answers were seen. Many students scored full or nearly full marks for this question.

5. [Maximum mark: 4] 25M.1.HL.TZ3.5

Construct a truth table for the following expression:

W = C OR NOT B AND NOT A OR B [4]

Markscheme

*Award **max** [4].*
Award [1] for every two correct rows.

A	B	C	W
0	0	0	**1**
0	0	1	**1**
0	1	0	**1**
0	1	1	**1**
1	0	0	**0**
1	0	1	**1**
1	1	0	**1**
1	1	1	**1**

Examiners report

There were a few truth tables with four inputs or eight inputs with duplications. Most responses to question 5 were correct. Some students did not recognise that logical AND operation has higher precedence than OR operation.

6. [Maximum mark: 2] 24N.1.SL.TZ0.8

Identify **two** advantages of low power consumption digital devices for users. [2]

Markscheme

*Award **[2 max]***

low power consumption prolongs battery life/No need to charge the device as often;
less energy consumption - cheaper energy bills;
increased mobility/portability of devices;
Device is less likely to overheat;

Examiners report

A vast majority of students earned one or two marks here. Those who lost marks typically reiterated their points by expressing them differently.

7. [Maximum mark: 2] 24N.1.SL.TZ0.1

State **two** features of spreadsheets. [2]

Markscheme

Award ***[2 max]***

Neatly organized information (rows and columns, tables, dropdown lists, worksheets)
Automatic calculation based on the formulas/functions (custom or pre-set);
Data filtering;
Data visualization (bar charts/pie charts/graphs/ variety of styles and colours);
Cell/table formatting (custom or pre-set) (sizes, colours, cell background, insert headers, merge cells, etc.);
Conditional formatting;
Automated tasks (auto-fill cells, update cell values, automated formatting);
Text manipulation;
Sorting;
Pivot tables;
Data validation;
Comments/ notes (for collaboration and clarification);
Sharing and co-authoring (multiple users can work on one document simultaneously);
Data import/ export (easily import data from various sources (e.g., databases), save spreadsheets in different formats (e.g., PDF);
Security features (password protection, file encryption);
Accessibility features (screen reader, assistive technologies for visually impaired users, keyboard shortcuts);

Note*: Reward other correct responses.*

Examiners report

SL:

Students who gave a more specific answer identifying the features of spreadsheet, scored better than the ones who gave generic answers such as 'performs calculations'.

HL:

Students were generally able to name or describe at least one feature of spreadsheets.

8. [Maximum mark: 15] 24N.1.SL.TZ0.11

Laptops are used daily for data storage, web browsing, gaming, and email.

(a) State **one** precaution a user can take to secure their data in case their laptop is stolen. [1]

Markscheme

Award ***[1 max]***

A password should be assigned to the device (biometric passwords could be used);
Should not store sensitive data on the laptop;
Safeguard all passwords / should not store username/password account logins (or "remember me" cookies) on the device;
Encrypt the SSD / hard drive;

Note*: Reward other correct responses.*

Note*: Backing-up will not keep the data secure, it will only allow recovery of data removed, so no marks for the answer: 'Regularly back up the SSD / hard drive to another location'.*

Examiners report

SL:

Several students incorrectly mentioned backup or cloud storage as a security method and lost marks here.

HL:

Students were mostly able to state one precaution a user can take to secure their data in case their laptop is stolen.

(b) Outline **one** feature of the operating system that is required when running a program on a laptop. [2]

Markscheme

*Award **[2 max]***
*Award **[1]** for identifying an OS feature and award **[1]** for a reasonable expansion.*

Security management;
confidential data stored in the system is protected by the operating system/ the system is protected from malware attack;

GUI / I/O operations;
OS can handle I/O operations to hide the behaviour of hardware from the user;

Process management;
The program execution is managed effectively by the operating system without any overlapping or time delay/The OS to develop and provides mechanism for communication and synchronization within multiple processes;

Storage/ Memory management;
OS performs memory management and virtual memory multitasking/ The need for memory management in OS is to allocate and de-allocate memory space to process/ The OS can do resource allocation and prevent the system from overloading /to ensure it meets the minimum requirements of the application;

Disk management;
OS permits disk access to manage files systems, file system device drivers and related activities of files like retrieval, naming, sharing, storage, protection of files;

Loading and execution;
The command interpretation is made to interpret the given commands and make the resources to act on the system by processing the commands;

Note: *Reward other correct answers*.

Examiners report

SL:

Most students gained a mark here for mentioning a feature of the operating system but failed to give a reasonable expansion. There were some who mentioned two features but did not expand on either of them thus gaining only one mark. Students should also be aware that at this level of study there is a need to demonstrate more than just general knowledge, and statements such as 'memory management manages memory' are far too vague for credit.

HL:

Students were generally able to state one feature of the OS that is required when running a program. Students who were able to provide a suitable expansion on this point achieved both marks.

(c) Justify the decision of the laptop manufacturer to include both wired **and** wireless network connection capability. [4]

Markscheme

*Award **[4 max]***

(the laptop manufacturer includes both wired and wireless network connection capability)
to deliver customer (laptop user/owner) satisfaction/ customer loyalty/ improve overall customer experience/ meet customer expectations;
laptop users (customers) can decide which connection is best (for them) in a particular situation;
users can move around freely / able to work in a setting outside the home/ office (and still stay connected from any location within the wireless network's coverage area or from any WIFI hotspot);
for tasks that require large amounts of data to be transferred users are able to use a wired network rather than wireless as the bandwidth is much larger/ faster transmission;
where wireless connection is weak/ unavailable users can count on an Ethernet cable adapter/ use wired network to get the connection they need;
wireless networks can be easily hacked, so wired connection can be used when working / transferring sensitive data/ it is more difficult for unauthorised users to intercept data in a wired network;

***Note**: Reward other reasonable benefits to their customers (laptop users) (knowing the advantages/ disadvantages of wired/ wireless connections) such as convenience, increased mobility, security, cost.*

Examiners report

SL:

Most students earned at least two or three marks here. They need to recognize that statements like "it does not use wires/cables" without further elaboration on its benefits are insufficient for gaining marks.

HL:

Generally, well answered, with many full or near full marks awarded.

A data packet is a basic unit of communication over a computer network.

(d) Describe the structure of a data packet. [2]

Markscheme

Award ***[2 max]***

The data packet structure includes the header, payload (and trailer);
It contains information about the data carried by the data packet such as its origin and destination IP addresses, number of packets, etc.;
the actual data (payload) that the packet is delivering to the destination;
and control signals/ additional information about the packet (such as data that tell the receiving device that it has reached the end of the packet);

Examiners report

SL:

Some unnecessary confusion was seen here. Responses included vague references to the process of packet switching rather than simply the structure of a packet.

HL:

The data packet structure includes the header, payload and trailer. Students mostly gave good responses for this question.

(e) Outline **three** reasons why protocols are necessary on a computer network. [6]

Markscheme

*Award **[6 max]***
*Award **[1]** for a reason, award **[1]** for an extension, **x3**.*

(Network protocols are necessary because protocols do)

provide the rules for effectively managing/ operating a computer network;
allowing network administrators to monitor network performance/ detect bottlenecks /identify and troubleshoot network issues/ configure network devices remotely / for example, network management protocols such as ICMP and SNMP;

define the policies/ procedures to determine how data is transmitted between different devices in the network;
for example, network communication protocols such as TCP/IP and HTTP;

provide security services (for example, encryption, authentication) for data transmitted over the network;
for example, network security protocols such as IPSec, HTTPS, SFTP, and SSL;

include mechanisms/functions for flow control;
mechanisms for devices to identify and make connections/ formatting rules that specify how data is packaged into messages sent and received / the amount of transferred data between the communicating computers must

be agreed in such a way that the data can always be stored on the target computer;

include error checking functions (in a communication protocol);
that help to detect errors/ eliminate distortions (for example, distortions that can be caused by the poor quality of the transmission, etc.);

include mechanisms for congestion control /include precautions that serve not to overload anetwork;
because when overloading a network, the transmitted data blocks often have to be discarded/ the transmission time of data blocks in the network by 'congestion' in nodes increases;

include mechanisms for deadlock control/ mechanisms to prevent/detect/avoid deadlock;
a situation that occurs when a process of transmission is in a wait state and some packets cannot advance toward their destination because are waiting on one another to release resources;

ensure integrity /accuracy/ completeness/ consistency/ validity of data;
by organizing data in a way that ensures the secure transmission between the origin/sender and destination/receiver;

Examiners report

SL:

High marks for this question were rare. General responses such as "to transfer data" without further explanation of how protocols are involved in the data transfer received no marks.

HL:

Students generally recognised that protocols define rules that govern network communication, provide the rules for effectively managing a computer network, provide security services, include mechanisms for flow control, congestion control and deadlock control, etc.

9. [Maximum mark: 15] 24N.1.SL.TZ0.12

(a) Consider the following expression:

(X > 6) OR (Y > 3) AND ((X + Y) < 20)

(a.i) State **all** the Boolean operators in this expression. [1]

Markscheme

Award ***[1 max]***

OR, AND;

Examiners report

SL:

Most students were able to score a mark. Some of them incorrectly mentioned relational operators instead of Boolean operators and lost the mark.

HL:

Mostly a well answered question.

Some students confused Boolean operators with relational operators.

(a.ii) State **all** the constants in this expression. [1]

Markscheme

*Award **[1 max]***

6, 3, 20;

Examiners report

SL:

Very few students were aware of the meaning of a 'constant' and confused it with a variable.

HL:

Some students confused constants with variables.

(a.iii) Determine the value of this expression when X is 6 and Y is 6. Show all your working. [2]

Markscheme

*Award **[2 max]***
*Award **[1]** for the correct result (True) and award **[1]** for working.*

(6 > 6) OR (6 > 3) AND (6 + 6 < 20)
False OR True AND True
False OR True
True

Examiners report

SL:

Most students gained at least one mark here for the final answer but lost marks for working by evaluating 'OR' before the 'AND'.

HL:

Most students were able to correctly determine the value of the given expression.

A student plans to create an alarm system for their room. The room has one door and one window.

Three sensors will be used in the system:

- A sensor to detect movement in the room.
- A sensor to detect if the door is locked or unlocked.
- A sensor to detect if the window is open or closed.

The student knows that this practical problem can be expressed in terms of Boolean logic and presented in a truth table.

The student considers the following three inputs:

1. MOTION, where true (1), represents that movement is detected in the room and false (0) represents no movement is detected.
2. DOOR, where true (1), represents that the door is locked and false (0) represents an unlocked door.
3. WINDOW, where true (1), represents that the window is open and false (0) represents a closed window.

A siren will make a warning noise when the door is locked and **either** of the following occur:

- Movement is detected.
- The window is open.

(b) Copy and complete the truth table for this alarm system.

MOTION	DOOR	WINDOW	SIREN
0	0	0	

[4]

Markscheme

*Award **[4 max]***

*Award **[1]** for every two correct rows.*

MOTION	DOOR	WINDOW	SIREN
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

Examiners report

SL:

Attempted well by most of the students though there were some who used ticks and crosses instead of 0's and 1's as inputs and lost marks. The table in the question clearly showed sample input and should have been followed through.

HL:

Students demonstrated that they could complete a truth table for a given problem, and as a result, achieved full, or near full, marks. A small number of students lost marks for errors such as an incorrect number of rows in the truth table or due to not systematically filling the input columns, but using a more random approach, they may have provided 8 input combinations, but at least one was a repeat of another.

The central processing unit (CPU) executes a program that is stored as a sequence of machine language instructions in primary memory. It does this by repeatedly fetching an instruction from primary memory, decoding that instruction, executing and storing the result.

(c.i) Identify the CPU register that holds data to be transferred to primary memory. [1]

Markscheme

*Award **[1 max]***

Memory Data Register/ MDR;

Examiners report

SL:

Some students confused this with MAR and lost a mark.

HL:

Most students answered this question reasonably well.

(c.ii) Identify the part of the CPU that performs decoding. [1]

Markscheme

*Award **[1 max]***

Control Unit/ CU;

Examiners report

SL:

Mostly attempted well.

HL:

A few students incorrectly answered that ALU performs decoding.

(c.iii) State where the calculations will be executed. [1]

Markscheme

*Award **[1 max]***

Arithmetic and Logic Unit/ ALU;

Examiners report

Attempted well by most of the students.

(c.iv) Explain the role of buses in the execution of a machine language instruction. [4]

Markscheme

*Award **[4 max]***

Buses connect hardware components/ transfer information/signals between different components (CU, ALU, RAM, registers);
Address bus carries the address of the memory location from where data should be read/ written:
Data bus carries the actual data which need to be stored on/ written to the primary memory;
Control bus carries control/ timing signals from CU to other components;
Buses carry power supply to all components;

Examiners report

Most students were able to earn two or three marks here. While many described the data bus and address bus, very few students also addressed the control bus.

10. [Maximum mark: 3] 24M.1.SL.TZ1.9

(a) Outline the purpose of the memory data register (MDR). [2]

Markscheme

*Award **[2 max]***

To hold the data / instructions;
to be transferred/fetched to / from the main memory / RAM of the computer OR that has been read or needs to be written;
using the address from the MAR;

Examiners report

Students had difficulty in expressing the purpose of MDR using correct keywords. Responses which stated that MDR "fetches" or "transfers" data did not score full mark.

(b) Outline the role of the arithmetic logic unit (ALU). [1]

Markscheme

Award ***[2 max]***

The ALU performs simple arithmetic operations / mathematical calculations / addition, subtraction, multiplication, division;
… and logical comparisons / AND, OR and NOT operations;

Examiners report

Majority of the students scored full marks by correctly outlining the role of ALU in performing arithmetic and logical operations.

11. [Maximum mark: 2] 24M.1.SL.TZ1.10

(a) State the binary equivalent of the denary number 37. [1]

Markscheme

Award ***[1 max]***

00100101 *OR* 100101 *OR* 0100101;

Examiners report

This question had the most correct answer. Students were able to convert the given denary number to binary.

(b) State the hexadecimal equivalent of the binary number 10011110. [1]

Markscheme

Award ***[1 max]***

9E;

Examiners report

Majority of the students were able to express the given binary number in hexadecimal format. However, some students did not apply the correct method and tried converting to denary incorrectly.

12. [Maximum mark: 2] 24M.1.SL.TZ1.11

Distinguish between **two** types of primary memory. [2]

Markscheme

Award ***[2 max]***

Award ***[1]*** *for each comparison up to* ***[2]***

RAM is a volatile / temporary memory (which could store the data as long as the power is supplied) and ROM is a permanent / non-volatile memory

(which could retain the data even when power is turned off);
Data stored in RAM can be altered whilst data stored in ROM can only be read;
RAM is used to store the instructions / data / programs that are currently being processed / in use whereas ROM is used to store permanent data / files such as start-up instructions for the computer.
RAM - large physical chip size / higher capacity / expensive whilst ROM - small size / less capacity / cheaper;
The CPU can access the data stored in RAM directly whilst the CPU cannot access the data stored on ROM (unless the data is stored in RAM) / whilst ROM data can be accessed after transferring it to RAM

Note: Do not award mark for Cache memory.

Examiners report

Some students did not identify the primary memory types correctly. Their responses included comparisons between hard disk drive, solid state drive, cache etc.

13. [Maximum mark: 15] 24M.1.SL.TZ1.14

(a) Define the NOR Boolean operator. [1]

Markscheme

*Award **[1 max]***
*Award **[1]** for either the description OR the truth table.*

Outputs the value of one if and only if all inputs have a value of zero;

A	B	Z
0	0	1
0	1	0
1	0	0
1	1	0

Note: DO NOT accept - reverse / negates

Examiners report

Many students have correctly defined the NOR operator in words. Some of them have written the equivalent truth table for which they achieved full mark. However, a few of them have just written "OR and NOT combination" which did not fetch any mark.

A car has features that monitor its speed, direction and distance from the car in front. This is shown in **Figure 1**.

Figure 1: Rules to control car motion

Input	Binary representation	Description
A	0	Car is less than 20 metres from the vehicle in front.
	1	Car is 20 metres or more from the vehicle in front.
B	0	Car is travelling in reverse or stationary.
	1	Car is travelling forward.
C	0	Car speed is more than 130 kilometres per hour.
	1	Car speed is 0–130 kilometres per hour.

For example, if the car is travelling forward, input B would have a binary representation of 1.

(b) Construct a logic diagram with inputs A, B, and C and output Z to represent the following scenario:

Output Z equals 1 when:

- the car is travelling forward AND it is less than 20 metres from the vehicle in front.
  OR
- the car speed is more than 130 km per hour.

In all other conditions, output Z equals 0. [4]

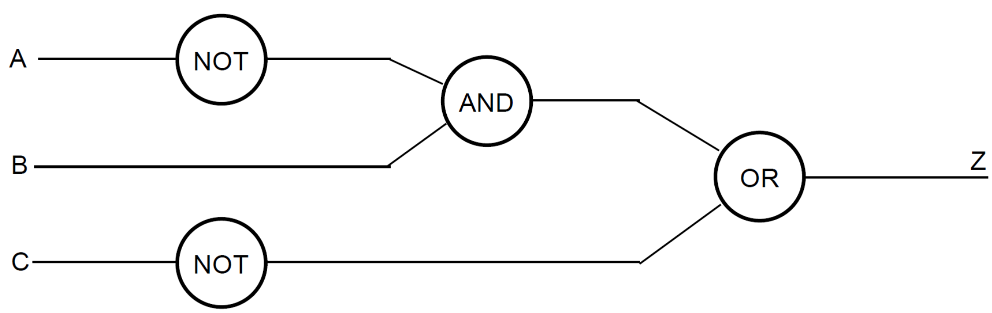
Markscheme

*Award **[4 max]***

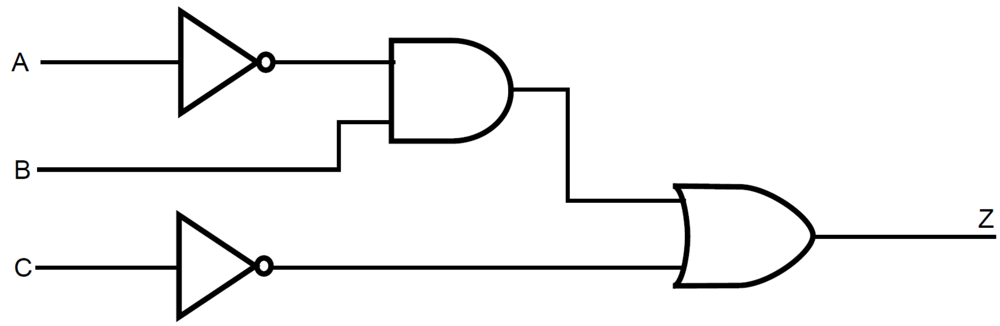
Correct NOT gate with input A and single output;
Correct NOT gate with input C and single output;
Correct AND gate with two inputs - one input from B and one output;
Correct OR gate with two inputs and one output Z;

Answer should represent Z = A′.B + C′

Example answer



Alternative answer



Examiners report

Many students have misread the question and constructed a truth table instead of producing a logical diagram. Students are recommended to use the correct number of inputs for each gate and also label the inputs and output.

An additional row (input D) is to be added to assist when the car is in reverse or stationary. Input D checks if there are obstructions less than 3 metres from the rear of the car.

(c) State the rules that need to be added to **Figure 1** to test this condition. [2]

Markscheme

*Award **[2 max]***

*Award **[1]** for each correct row (The rules for 0 and 1 can be reversed).*

D	0	Obstruction is less than 3 metres from the rear of the car / There are obstruction less than 3 metres from the rear of the car
	1	Obstruction is 3 or more metres from the rear of the car / There are no obstructions less than 3 metres from the rear of the car

Examiners report

While about 50% of the students could correctly state the rules, the other 50% did not fully specify the conditions for input D.

Information similar to that presented in **Figure 1** could be used to construct decisions and conditions in program design (see **Figure 2**).

Figure 2: Identifiers for car motion rules

Identifier	Description
F	Distance in metres to the vehicle in front
S	Speed of car in kilometres per hour
T	Travelling in a forward direction

(d) Determine the value of the following expression given that the input values for F, S and T are:

F = 5
S = 30
T = true

F >= 25 AND S >= 10 AND S <= 130 AND T = true

You must show your working. [2]

Markscheme

*Award **[2 max]***
*Award **[1]** for evidence of working e.g. substitution into variables to evaluate the expression;*
*Award **[1]** for correct answer false;*

Example answer

5 >= 25 AND 30 >= 10 AND 30 <= 130 AND true
false AND true AND true AND true
(Output =) false

Note: Accept 1/0 instead of TRUE/FALSE

Examiners report

This question had most of the students who attempted getting full marks. A very few students did not show the working properly breaking down the conditions in a stepwise manner.

(e) Construct an algorithm in pseudocode that repeats the following steps while the car is moving:

- Input the value for the distance from the vehicle in front.
- Input the value for the speed of the car.
- Check the inputs and notify the user if either the distance from the car in front is less than 20 metres or if the speed of the car is more than 130 kilometres per hour.

The algorithm will only terminate when the car stops moving. [6]

Markscheme

Award ***[6 max]***

Use of flag (or otherwise) to maintain continuous loop;
Appropriate loop structure – while / repeat until;
Inputs for distance and speed inside the loop;
Correct conditions to implement alarm notification;
Correct Output of alarm;
Correct condition to cause loop to stop;

***Note**: Do not accept break to terminate the loop*

Example answer 1

```
FLAG = true
loop while FLAG
    input D // allow read D
    input S // allow read S
    if D < 20 OR S > 130 then
        output "ALARM!!"// allow any other output
    end if
    if S = 0 then
        FLAG = false
    end if
end loop
```

*Award **[6 max]***

Input of speed before the loop;
Appropriate loop structure – while / repeat until;
Inputs for distance and speed - after the endif. Both inside the loop;
Correct condition to implement alarm notification;
Correct Output of alarm;
Correct condition to cause loop to stop;

Example answer 2

```
input S
loop while S > 0
   input D // allow read D
   if D < 20 OR S > 130 then
      output "ALARM!!"
```

```
    end if
    input S // allow read S
end loop
```

Examiners report

Students who constructed algorithms using a loop to take the inputs and also devised a way to stop the loop using correct condition achieved full marks. Students need to use correct output statements and consider the steps that need to be present inside and outside the loop carefully. They are reminded to use pseudocode conventions provided.

14. [Maximum mark: 3] 23N.1.SL.TZ0.5

Construct a truth table for the following expression:

A OR NOT B AND C [3]

Markscheme

*Award **[3 max]**.*

*Award **[1]** for 5 correct rows in the truth table*

*Award **[2]** for 6 or 7 correct rows in the truth table*

*Award **[3]** for all 8 correct rows in the truth table*

A	B	C	A OR NOT B AND C
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Examiners report

Most students were able to score some marks for drawing the truth table for the given logic expression. Students who evaluated the operation in the right order were able to score full marks.

15. [Maximum mark: 2] 23N.1.SL.TZ0.7

Identify **three** functions of the control unit (CU) in the central processing unit (CPU). [2]

Markscheme

*Award **[3 max]**.*

Fetches/extracts each instruction from memory;
Decodes/transforms them into several commands/signals/steps (that are passed to the ALU or I/O or other components in the CPU for execution);
Controls the movements of data within the CPU;
Generates the clock pulses that regulates speed of the instruction cycle;
Generates control signals for all hardware components to regulate their activities;
Synchronizes all the operations of the CPU;

Note: Accept other reasonable answers.

Examiners report

Students who wrote specific answers referring to data, memory, instructions etc. achieved better marks than those who wrote general answers such as 'CU decodes and executes'.

16. [Maximum mark: 1] 23N.1.HL.TZ0.2

Define the Boolean NAND operator. [1]

Markscheme

*Award **[1 max]**.*

The NAND operator produces a FALSE value only if (and only if) both values of its two inputs are TRUE;

Note: Accept a correct truth table for the expression A NAND B.

Examiners report

Many students were able to define the Boolean NAND operator.

17. [Maximum mark: 2] 23M.1.SL.TZ1.1

Outline the function of a web browser. [2]

Markscheme

*Award **[2 max]***

web browser allows users access to information/resources on the WWW;
(when a user asks for a particular website) the web browser fetches the required content from a web server/ acts as an interface between a client and server;
prepares the retrieved information to be displayed/ interprets the content to be rendered in a format that can be understood/ displays the resulting web page on the user's device;
allows user to navigate around website/ open more than one web page/ print/ save page/ etc.;

Examiners report

SL:

Candidates who were able to outline the function of a web browser by relating its use to the World Wide Web or its interaction with a web server achieved the best marks.

HL:

Most candidates were able to outline the function of a web browser.

18. [Maximum mark: 4] 23M.1.SL.TZ1.4

(a) Identify **one** characteristic of random access memory (RAM). [1]

Markscheme

*Award **[1 max]***

Random Access Memory is volatile / requires power to maintain the stored information/ data is retained in RAM as long as the computer is on, but it is lost when the computer is turned off;

Data/instructions that are currently being used are stored in RAM;

RAM can be modified, erased, or read/ data stored in RAM can be altered;

RAM is small in capacity compared to secondary storage media;

It is fast (faster to read from and write to than other kinds of storage, such as a hard disk drive (HDD)/ solid-state drive (SSD));

Examiners report

Most successful candidates identified the fact that RAM is volatile as a characteristic. Other similar responses were also seen.

(b) Explain the use of cache memory. [3]

Markscheme

*Award **[3 max]***
Cache memory is (an extremely) fast memory type;
that acts as a buffer between RAM and the CPU;
it holds frequently requested data and instructions;
so that they are immediately available to the CPU when needed;

Cache memory is used as an intermediate form of storage between very high-speed CPU registers and the slower RAM;
located inside the CPU and can be directly accessed by the processor;
It is used to store instructions and data that are repeatedly required during the execution of programs; thus, improving the performance and speed of the whole system/ thus avoiding the need to access the dynamic RAM to retrieve the same data repeatedly/ is used to reduce the average time to access data from the main memory;

Examiners report

Most successful candidates were able to explain or partially explain that cache is a very fast type of memory used to store frequently used data and instructions, going on to indicate that this helped to speed up the operation of the whole system.

19. [Maximum mark: 4] 23M.1.SL.TZ1.5

Construct a truth table for the logic expression

A NAND (B NOR C) [4]

Markscheme

*Award **[4 max]***

*Award **[1]** for every 2 correct rows in the truth table.*

A	**B**	**C**	**A NAND (B NOR C)**
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Examiners report

SL:

Most candidates were able to score some marks for drawing the correct truth table for the given logic expression. A wide range of marks were seen, with many high scoring candidates.

HL:

Many candidates scored full or nearly full marks for constructing a truth table for the logic expression

(A NAND B) NOR C

20. [Maximum mark: 2] 23M.1.HL.TZ1.3

Outline **one** reason for using Unicode to represent data in a computer system. [2]

Markscheme

Award ***[2 max]***

Because Unicode is an (established) standard for data representation/ a single encoding scheme for all languages and characters;
So, data can be used/transported through many different systems/ platforms/ devices;

Because of the difference between ASCII and Unicode in the number of bits used to encode (ASCII (8-bits) and Unicode (16/32 bits));
It can represent over a million characters/ ASCII cannot be used to encode the many types of characters found around the world;

Because Unicode can be defined with different character encoding like UTF-8, UTF-16, UTF-32, etc.;
And is used to represent many (over a million) characters from many (more than 150) modern and historic scripts (along with emoji);

Examiners report

SL:

Candidates whose responses outlined that Unicode is an established standard for data representation, or who were able to outline the relative higher number of characters capable of being stored, compared with ASCII, achieved the highest marks.

HL:

A good range of answers was seen for this question. Many answers were excellent. A small number of candidates confused Unicode with a programming language.

21. [Maximum mark: 2] 23M.1.HL.TZ1.7

Distinguish between random access memory (RAM) and read-only memory (ROM). [2]

Markscheme

*Award **[2 max]***
*Award **[1]** for each comparison up to **[2]**.*

RAM is a volatile memory (which could store the data as long as the power is supplied) whilst ROM is a non-volatile memory (which could retain the data even when power is turned off);
Data stored in RAM can be altered whilst data stored in ROM can only be read;
RAM is used to store the data that has to be currently processed by CPU whilst ROM stores the instructions required during bootstrap of the computer;
RAM- large physical chip size/ higher capacity/expensive whilst ROM- small size/ less capacity/cheaper;
The CPU can access the data stored in RAM whilst the CPU cannot access the data stored on ROM (unless the data is stored in RAM);

Examiners report

A well answered question. Many candidates were able to distinguish between RAM and ROM.

22. [Maximum mark: 2] 22N.1.SL.TZ0.7

Each pixel on a computer screen has three colour values associated with it: red, green and blue. The range for each of the three colour values is from $0_{(10)}$ to $255_{(10)}$.

Colour values can also be represented in hexadecimal. For example, the colour blue can be represented in hexadecimal as 0000FF.

(a) State the binary representation of the colour blue. [1]

Markscheme

Award ***[1]*** *max.*

000000000000000011111111;

Examiners report

The vast majority of candidates were able to state the binary equivalent of the hexadecimal 0000FF.

(b) State the number of colours that can be represented in each pixel on the computer screen. [1]

Markscheme

Award ***[1]*** *max.*

2^{24} / $(2^8)^3$ / 256^3;

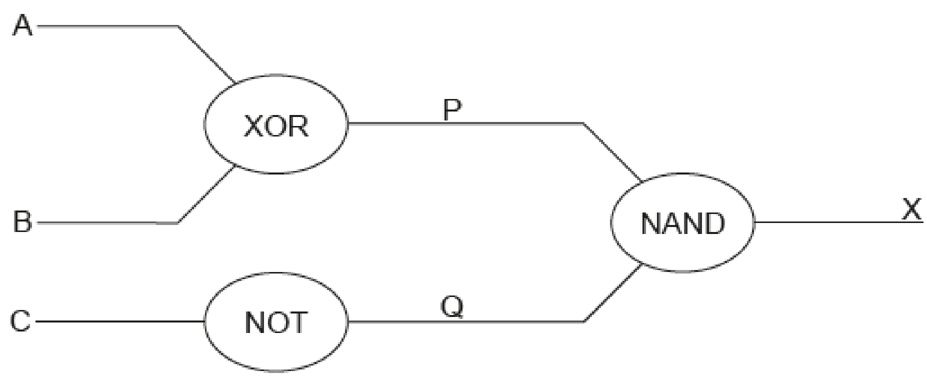
16.8 million / 16,777,216;

Examiners report

A wide range of responses was seen for this question, with a relatively small number of candidates achieving the correct response of 16.8 million colours, or its equivalent.

23. [Maximum mark: 4] 22M.1.SL.TZ0.6

Draw the truth table for the following logic circuit.



[4]

Markscheme

*Award **[4 max]**.*

1 mark for every two correct rows;

A	B	C	P	Q	X
0	0	0	0	1	1
0	0	1	0	0	1
0	1	0	1	1	0
0	1	1	1	0	1
1	0	0	1	1	0
1	0	1	1	0	1
1	1	0	0	1	1
1	1	1	0	0	1

Examiners report

The majority of candidates achieved high marks for drawing and completing an appropriate truth table for the given logic circuit. Some marks were lost by candidates not including all possible combinations of inputs in their responses, or missing out some of the interim stages, therefore giving inaccurate outputs.